ABHAY PAREEK

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Summary

Recent B.Tech graduate in Electronics and Communication Engineering with hands-on experience in **Python**, **SQL**, **Power BI** and **Streamlit**, and in building and testing **LLM**-based tools. Skilled in **data cleaning**, **exploratory analysis** and **dashboard development**, with a focus on checking outputs and documenting results.

Technical Skills

Programming: **Python**, **SQL**
Data Analysis: **Pandas**, **NumPy**, **Data Cleaning**, **Exploratory Data Analysis**, **Statistical Analysis** (**Wilson intervals**, **Fisher exact test**)
Generative AI and LLMs: **LLM** integration (**Groq API**, **gpt-oss-20b**), **prompt design**, **structured output validation**, **text-to-SQL**, **AI evaluation**
Visualization and BI: **Power BI**, **DAX**, **Power Query**, **Streamlit**, **Matplotlib**
Databases: **PostgreSQL**, **SQLite**
Tools and Testing: **unittest**, **pytest**, **Git**, **GitHub**, **Jupyter**, **OpenPyXL**, **Microsoft Excel**

Projects

Recruitment Funnel and Source-of-Hire Analytics *Python, SQL, Power BI, Streamlit, Generative AI*
GitHub: github.com/22ec016abhay/recruitment-funnel-dashboard Live app: recruitment-funnel-dashboard.streamlit.app

- Built an end-to-end **pipeline** on 1,500 synthetic applicant records: cleaned source labels, loaded **SQLite** (3,078 funnel events), and built a **Power BI** report and a **Streamlit** app.
- Analysed **hire rates** by source and stage: Referral 19.52% vs Job Fair 2.65%, overall hire rate 7.87%, average **time-to-hire** 35.46 days.
- Built a **weekly brief generator** with an **LLM**: the model writes a template, Python fills in every figure, and a **validator** checks each number against the data. Documented failed runs alongside successful ones.
- Built a **text-to-SQL** assistant with a **read-only** query guard, row and time limits, and refusal rules. Scored 12 of 15, then 15 of 15, on answerable questions across two runs.
- Ran a gender selection-rate analysis (**Wilson intervals**, **Fisher exact tests**, 4/5ths ratios) on synthetic data with a deliberately planted gap, to test whether the method detects it.

Customer 360: E-commerce Growth and Retention Intelligence *Python, SQL, PostgreSQL, Power BI*
GitHub: github.com/22ec016abhay/Customer-360-ecommerce-intelligence

- Built an e-commerce **data pipeline** in **Python** (**Pandas**, **NumPy**) for validation, **data cleaning** and outlier treatment across large order records.
- Implemented **RFM segmentation** to categorize customers into Champions, Loyal, At Risk and Lost segments.
- Ran **PostgreSQL** analysis (CTEs, window functions, GroupBy) on sales concentration (Pareto principle), repeat purchase rates and product profit margins.
- Designed a 7-page **Power BI** dashboard with drill-throughs, custom tooltips and what-if scenarios; found At Risk customers accounted for 34% of total revenue opportunity.

RavenStack: Subscription Health and Revenue Retention Analytics *SQL, PostgreSQL, Power BI, Excel*
GitHub: github.com/22ec016abhay/ravenstack-subscription-analytics

- Analysed **churn** and retention across about 500 subscription accounts, building **KPI** measures for MRR, churn rate and a customer risk score.
- Found churn concentrated by industry and geography (DevTools at 30.97%, Germany at 32%) rather than product usage or support experience.
- Found monthly-billing customers over-represented among at-risk accounts, and recommended a targeted incentive to move them to annual billing.
- Found the existing rule-based risk model under-weights industry and geography, the strongest churn signals.

InsightFlow Analytics: End-to-End Retail Data Pipeline *Python, PostgreSQL, Pandas, Matplotlib, OpenPyXL*
GitHub: github.com/22ec016abhay/InsightFlow-Analytics

- Built a retail **pipeline** processing 15,000+ orders across 15 stores and 2,000 customers, from raw CSV ingestion through validation and analysis.
- Designed a normalized **PostgreSQL** schema with keys and indexes, and a **Python** connection layer using context managers.
- Cleaned data in **Pandas** and **NumPy** and generated a multi-sheet **Excel** report (**OpenPyXL**) and charts (**Matplotlib**), validated with 19 **pytest** unit tests.

Certifications

- **GenAI Powered Data Analytics** Job Simulation, Tata (Jul 2026)
- **Data Science** Job Simulation, Lloyds Banking Group (Oct 2025)
- **Data Analytics** Job Simulation, Quantium (Sep 2025)
- **Data Analytics** Job Simulation, Deloitte Australia (Sep 2025)

Education

B.Tech, Electronics and Communication Engineering, JSS Academy of Technical Education, Noida (Jun 2026)